

Exercises after classes 3-4

Exercise 1. Calculate the variance of random variables X and Y in the schoolbus problem.

Exercise 2. With a fair coin what is the probability of getting s heads before t tails?

Exercise 3. We have a possibly biased coin. Design an event whose probability is $1/2$.

Exercise 4. We have a fair coin. Design an event whose probability is $1/3$.

Exercise 5. We have a possibly biased coin. Design an event whose probability is $1/5$.

Exercise 6. You have \$1000, and a certain commodity presently sells for \$2 per ounce. Suppose that after one week the commodity will sell for either \$1 or \$4 an ounce, with these two possibilities being equally likely.

- (a) If your objective is to maximize the expected amount of money that you possess at the end of the week, what strategy should you employ?
- (b) If your objective is to maximize the expected amount of the commodity that you possess at the end of the week, what strategy should you employ?

Exercise 7. To determine whether they have a certain disease, 100 people are to have their blood tested. However, rather than testing each individual separately, it has been decided first to place the people into groups of 10. The blood samples of the 10 people in each group will be pooled and analyzed together. If the test is negative, one test will suffice for the 10 people, whereas if the test is positive, each of the 10 people will also be individually tested and, in all, 11 tests will be made on this group. Assume that the probability that a person has the disease is .1 for all people, independently of each other, and compute the expected number of tests necessary for each group. (Note that we are assuming that the pooled test will be positive if at least one person in the pool has the disease.)